

Course Code	Course Name	Credits	Syllabus	References
CS6140	Video content analysis	3	Introduction to video content analysis, feature extraction, video structure analysis –shot and scene segmentation, content based video classification, video abstraction – skimming and summarization, event detection and classification, indexing for retrieval and browsing, Applications –Movie and sports video analysis, news video indexing and retrieval etc.	1. Video Content Analysis using Multimodal information, Ying Li, Kuo, C.C. Jay, Springer, 2003. 2. The Essential Guide to Video Processing, Al Bovik, Second Edition, Elsevier Academic Press, 2009. 3. Handbook of Image and Video Processing, Al Bovik, Second edition, Elsevier Academic Press, 2005. 4. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2010
CS6160	Cryptology	3	Mathematical foundations, Classic cryptosystems, perfect secrecy, private and public key cryptosystems and their cryptanalysis, secret sharing schemes and zero knowledge proofs.	1. Cryptography - Theory and Practice 2nd or 3rd Edition by Douglas Stinson. 2. An introduction to number theory and cryptography - Neal Koblitz. 3. "The Code Book" by Simon Singh and "The Codebreakers" by David Kahn both chronicle the history of cryptography. 4. A. Menezes, P. C. Van Oorschot & S. A. Vanstone: Handbook of Applied Cryptography,
CS6170	Computer Vision for Autonomous Vehicle Technology	3	vehicles. We will describe essential computer vision techniques for scene understanding, obstacle detection, semantic segmentation, lane detection and prediction, traffic-light & traffic-sign detection, multi-object tracking (MOT), trajectory switching, detection of constructions on the road, occlusion of vehicles, pedestrian intention estimation, road-agent behavior prediction, depth estimation from 2D images, road-scene analysis in adverse weather conditions, vulnerable road user (VRU) pose estimation, road segmentation. We will also briefly describe techniques related to real-time processing, model optimization.	Research papers
CS6190	Advanced Topics in Cryptology	3	Reading research papers in the area of cryptology and understanding the state of the art in the subject.	Research papers

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CS6200	Advanced topics in formal methods	3	This course will involve a reading of important papers in the area of formal methods. It will be preceded by a review of prerequisite concepts in logic, verification, model checking and automata theory.	1. Logic in Computer Science; Michael Huth and Mark Ryan; Cambridge University Press 2. Automata Theory and its Applications; Bakhadyr Khoussainov and Anil Nerode; Birkhauser 3. Algebraic Automata Theory; WML Holcombe; Cambridge Studies in Advanced Mathematics 4. Mathematical Foundations of Automata Theory (textbook preprint); Jean-Eric Pin
CS6210	Advanced machine learning	3	Generative models for discrete data, Gaussian Models, Bayesian Statistics, Linear Regression, Logistic Regression, Directed graphical models (Bayes nets), Mixture models and the EM algorithm, Sparse linear models. Kernels: Kernel functions, kernel trick, Support vector machines (SVMs), Kernels for building generative models. Markov and hidden Markov models, State space models, Undirected graphical models (Markov random fields), Monte Carlo inference, Markov chain Monte Carlo (MCMC) inference, Graphical model structure learning, Deep learning, Boosting, On-Line learning, Decision Trees, Ranking. Compressive Sensing and Dictionary Learning: Pursuit algorithms and applications for imaging and vision.	1. Machine Learning: a Probabilistic Perspective, Kevin P. Murphy, MIT Press, 2012. 2. Foundations of Machine Learning, Mehryar Mohri, Afshin Rostamizadeh and Ameet Talwalkar, MIT Press, 2012. 3. Introduction to Machine Learning, Ethem Alpaydin, 2nd Ed., MIT Press, 2009 4. Machine Learning: The Art and Science of Algorithms that make sense of Data, Cambridge
CS6220	Topics in networks	3	This hands-on research based advanced course will be primarily based on recent advances in core computer networking technologies and/or mobile wireless networks like Software Defined Networking, Future Internet, Cloud RAN, 5G, Convergence of Heterogeneous wireless technologies, M2M/IoT, etc. The objective is to discuss in depth some of the key advances from the literature, propose extensions, implement them on a testbed/NS-3 platform, and publishing the results.	Research papers, NS-3 network simulator.

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CS6230	Optimization methods in machine learning	3	Introduction to Optimization, Convex Sets, Convex Functions, Lagrange Duality, Convex Optimization Algorithms, Second-order cone models, Semi-definite programming, Semi-infinite programming, Minimax, Sublinear algorithms, Interior Point Methods, Active set, Stochastic gradient, Coordinate descent, Cutting planes method, Applications to Image/Video/Multimedia Processing	1. Sra, Suvrit, Sebastian Nowozin, and Stephen J. Wright, eds. Optimization for machine learning. Mit Press, 2012. (ISBN: 9780262016469): https://mitpress.mit.edu/books/optimization-machine-learning 2. Roberto Battiti, Mauro Brunato. The LION Way: Machine Learning plus Intelligent Optimization. Lionsolver, Inc. 2013. (http://www.lionsolver.com/LIONbook/) 3. Bubeck, Sébastien. "Theory of Convex Optimization for Machine Learning." arXiv preprint arXiv:1405.4980, 2014. (http://www.princeton.edu/~sbubeck/Bubeck14.pdf)
CS6240	Advanced compiler design		The course will focus lexical analysis, syntactic analysis, semantic analysis, abstract syntax tree and code-generation as well as basic optimizations. The initial part of the course will focus on the classic techniques of lexical analysis and scanning/screening, syntactic analysis like bottom-up and top-down parsing techniques, semantic analysis, type-checking, abstract syntax tree and code generation. The latter part will focus on intermediate representations and simple optimizations like register allocation and instruction scheduling. This is a sister course to the CS3020/3021 being offered to B.Techs. A large focus of these courses would be on implementing parts of compiler for a subset of C++/Java. The M.Tech course would have an additional paper presentation and some research component.	1. Compilers: Principles, Techniques, and Tools ("Dragon book") by Alfred V. Aho (Author), Ravi Sethi (Author), Jeffrey D. Ullman 1986 2. Compiler Design "Syntactic and Semantic Analysis" by ReinHard Wilhelm, Helmut Seidl and Sebastian Hack, 2013 3. Advanced Compiler Design and Implementation by Steven Muchnick, 1997
CS6250	Advanced compiler optimizations		This is the sister course for CS5260 "Compiler Optimizations" being offered this semester. CS6250 is being floated for M.Techs and PhDs, while CS5260 is being floated for B.Techs. CS6250 would have additional paper readings, writing reviews/critiques of papers, as well as additional research component.	1. Compilers: Principles, Techniques, and Tools ("Dragon book") by Alfred V. Aho (Author), Ravi Sethi (Author), Jeffrey D. Ullman 1986 2. Advanced Compiler Design and Implementation by Steven Muchnick, 1997 3. Handbook of Compiler Optimizations by Priti Shankar and Y.N. Srikant 4. Optimizing Compilers for Modern Architectures: A Dependence-based Approach by Randy Allen, Ken Kennedy, 2001

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CS6260	Deep Learning in Healthcare	3	Part I: Medical image detection, medical image segmentation, medical image classification, medical image enhancement using deep learning Part II: Improving the performance of deep CNNs in medical image segmentation with limited resources, Deep active self-paced learning for biomedical image analysis, Anatomical-landmark-based deep learning for Alzheimer's disease diagnosis with structural magnetic resonance imaging, Multi-scale deep convolutional neural networks for emphysema classification and quantification, Opacity labeling of diffuse lung diseases in CT images using unsupervised and semi-supervised learning using deep learning. Part III: Residual sparse auto-encoders for unsupervised feature learning and its application to HEP-2 cell staining pattern recognition, computer-Aided diagnosis system in medical imaging, System overview and Clinical applications using deep learning.	(1) Chen, Yen-Wei, and Lakhmi C. Jain, eds. Deep Learning in Healthcare: Paradigms and Applications. Vol. 171. Springer Nature, 2019. (2) Goodfellow, Ian, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. Deep learning. Vol. 1, no. 2. Cambridge: MIT press, 2016.
CS6260	Topics in wireless networks	3	Research based course primarily based on recent advances in mobile wireless networks like Mobile Converged Networks, Cloud RAN, SDN, M2M/IoT, etc. The objective is to propose extensions, implement them on a testbed/NS-3 platform, and publish the results.	Research papers, NS-3 network simulator, OpenAirInterface. http://www.nsnam.org/ http://www.openairinterface.org/
CS6270	Topics in Database Management Systems	3	Research based course primarily based on recent advances in Databases such as data mining, big data management and analytics, scalable data management and computing, parallel and distributed databases, spatial and temporal data management, data warehousing, advanced query processing, etc. All students will have to present a research paper of their choice, either from a given list of papers or other papers subject to instructor's approval. There will be two exams (midsem/endsem) and a course project. The project is mandatorily an implementation oriented project with potential to be a publishable paper.	For background only: 1. Database Systems Concepts, A. Silberschatz, H. Korth and S. Sudarshan, McGraw Hill, 6th edition 2. Introduction to Database Systems, R. Ramakrishnan and J. Gehrke, 3rd edition

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CS6280	Topics in program analysis	3	This advanced course is about reading program analysis papers, or studying the existing tools for static analysis, abstract interpretation and verification. Some of the topics of interest are abstract domains (like Polyhedra, Octagons and Intervals), Linear ranking functions and termination algorithms of loop programs, Transitive closure of Affine Control Loops etc. Examples of tools that would be studied are Astree/Astrea from INRIA/ENS, RankFinder from TU-Munich, Terminator from MSR-Cambridge, LoopKiller/iRankFinder from Israel/MPI, Rank from ENS-Lyon, Yogi, Q Program Verifier, Corral from MSR-India. A special focus would be on various varieties of solvers: SAT based, LP/ILP based, Satisfiability Modulo theories (SMT) based, Reachability Modulo Theories (RMT) based.	CS6280 Topics in Program Analysis 3 CS3020 (Principles of Compiler Design) or their equivalent. Prior approval of the instructor is needed. This advanced course is about reading program analysis papers, or studying the existing tools for static analysis, abstract interpretation and verification. Some of the topics of interest are abstract domains (like Polyhedra, Octagons and Intervals), Linear ranking functions and termination algorithms of loop programs, Transitive closure of Affine Control Loops etc. Examples of tools that would be studied are Astree/Astrea from INRIA/ENS, RankFinder from TU-Munich, Terminator from MSR-Cambridge, LoopKiller/iRankFinder from Israel/MPI, Rank from ENS-Lyon, Yogi, Q Program Verifier, Corral from MSR-India. A special focus would be on various varieties of solvers: SAT based, LP/ILP based, Satisfiability Modulo theories (SMT) based, Reachability Modulo Theories (RMT) based. Background: Abstract Interpretation by P. Cousot and R. Cousot. http://www.di.ens.fr/~cousot/AI/IntroAbsInt.html Neilsen, Neisen and Hankin. Principles of Program Analysis. Weakly Relational Numerical Abstract Domains. PhD thesis, Antoine Mine. P. Cousot. Abstract Interpretation course at MIT. http://web.mit.edu/afs/athena.mit.edu/course/16/16.399/www/Research papers.

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CS6290	Algorithmic Techniques for large graphs	3	Sampling, sketching, and sparsification on graphs. Graph streaming – graph semi-streaming algorithms, estimating graph measures, lower bounds on streaming algorithms. Recent publications on the topics.	(1) Survey by Andrew McGregor (https://people.cs.umass.edu/mcgregor/papers/graphsurvey.pdf) (2) “Sketching as a Tool for Numerical Linear Algebra”, David P. Woodruff, Now Publishers, 2014 (3) “Communication Complexity (for Algorithm Designers)”, Tim Roughgarden, Foundations and Trends in Theoretical Computer Science, 2016
CS6300	Topics in compiler optimizations	3	This advanced graduate level course will focus on a melange of selected topics in Compiler Optimizations. It is mostly a research based course where the registrants will focus on studying state-of-the-art algorithms, in a traditional setting or in the polyhedral compilation: studying and improving the existing algorithms published in top compiler conferences or the ones implemented in LLVM, Polly, PPCG, Pluto, etc.	Text books and references Research papers published in PLDI, POPL, CGO, PACT, PPOPP, TOPLAS in the past couple of years. The LLVM, Polly, PPCG, Pluto compiler frameworks
CS6310	Quantum Computing 1	1	Introduction to Quantum Mechanics--the mathematics and physics; Quantum Circuits; Deutsch and Deutsch Jozsa algorithms	Quantum Computation and Quantum Information: Nielsen and Chuang, MIT Press
CS6320	Quantum Computing 2	1	Quantum Algorithms: Shor's Integer Factoring, Grover's unordered search, Hidden Subgroup Problem for various groups, Other Quantum Algorithms	Quantum Computation and Quantum Information: Nielsen and Chuang, MIT Press
CS6330	Quantum Computing 3	1	Quantum Error Correction, Quantum Information Theory and Quantum Cryptography	Quantum Computation and Quantum Information: Nielsen and Chuang, MIT Press
CS6340	Topics in quantum computing	3	Topics in quantum computing and information; Latest advances in the field.	Quantum Computation and Quantum Information: Nielsen and Chuang, MIT Press
CS6350	Topics in combinatorics	3	This advanced graduate level course on combinatorics will focus on selected topics such as extremal combinatorics, probabilistic techniques, algebraic method in combinatorics etc.	The Probabilistic Method, by N. Alon and J. H. Spencer, 3rd Edition, Wiley, 2008. Linear algebra methods in combinatorics, by L. Babai and P. Frankl Extremal Combinatorics with Applications in Computer Science, by S. Jukna (Springer 2001).

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CS6360	Advanced topics in machine learning	3	This advanced graduate level course on machine learning will focus on selected topics such as deep learning, probabilistic graphical models, optimization in machine learning, etc. The course assumes that the student has basic knowledge in machine learning, and will have a research focus. The objective of the course will be to get a deeper understanding of machine learning algorithms, especially those that are highly relevant for contemporary real-world applications.	Deep Learning, MIT Press, Yoshua Bengio et al. http://www.iro.umontreal.ca/~bengioy/dlbook/version-07-08-2015/dlbook.html Neural Networks and Deep Learning, Free online book. http://neuralnetworksanddeeplearning.com/ Optimization for Machine Learning, Suvrit Sra et al, MIT Press. https://mitpress.mit.edu/books/optimization-machine-learning
CS6370	Information retrieval	3	DB vs IR, Tokenization, Indexing, Representing terms and documents, Scoring models, Query-independent ranking, Learning to Rank, Evaluation in IR, Advanced Topics from Research Papers	[Book] "Introduction to Information Retrieval" by Manning, Raghavan and Schütze. [Book] "Search Engines: Information Retrieval in Practice" by Donald Metzler, Trevor Strohman, and W. Bruce Croft.
CS6383	Introduction to compiler engineering	1	Analyses/Transformations in LLVM. Methods of adding new FrontEnds and BackEnds to LLVM. Introduction to Pass-manager of LLVM. Adding new passes. Details of BackEnds.	Getting Started with LLVM Core Libraries, Packt Publishing Limited. http://www.amazon.in/Getting-Started-LLVM-Core-Libraries/dp/1782166920 LLVM Essentials http://www.amazon.com/LLVM-Essentials-Suyog-Sarda/dp/1785280805 The LLVM Compiler Infrastructure, http://llvm.org/
CS6393	Advanced compiler engineering	3	The LLVM Compiler infrastructure has been well accepted to be a well-settled infrastructure. In recent times, industries have proposed, and are working on many "post-LLVM" frameworks. These include compilers for Machine Learning (like MLIR and TVM) or compiler infrastructures for Network applications (like P4C and P4LLVM). In this course, we will study some of these systems: both at the theoretical level, as well as the engineering level. This course will focus heavily on the engineering and implementation aspects of these modern compiler infrastructures (like MLIR, Polly) and the underlying libraries they use (like ISL) and will involve substantial system building. Assignments could involve posting patches to these compilers.	1. Chris Lattner, Mehdi Amini, Uday Bondhugula, Albert Cohen, Andy Davis, Jacques Pienaar, River Riddle, Tatiana Shpeisman, Nicolas Vasilache, Oleksandr Zinenko, MLIR: A Compiler Infrastructure for the End of Moore's Law https://arxiv.org/abs/2002.11054 2. https://www.tensorflow.org/mlir 3. Multi-Level Intermediate Representation Overview https://mlir.llvm.org/ 4. Polly: LLVM Framework for High-Level Loop and Data-Locality Optimizations https://polly.llvm.org/ 5. Integer Set Library http://isl.gforge.inria.fr/

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CS6400	Constraint solving	1	This course will attempt to study the underlying techniques employed by modern day constraint solvers. In particular, solving techniques behind SAT, MaxSAT, Pseudo-Boolean constraint solving will be studied. In addition, this course will also attempt to take a look at SMT (Satisfiability Modulo Theories) solving.	1. Selected research papers will be made available to the students by the instructor. 2. Handbook of Satisfiability edited by A. Biere, M. Heule, H. Van Mareen and T walsh. ISO Press, 2009. ISBN (print) : 978-1-58603-929-5, ISBN (online): 978-1-60750-376-7. 3. Handbook of Constraint Programming, edited by F. Rossi, P. Van Bree and T. walsh. Elsevier Science, 2006. ISBN:978-0-44452-726-4
CS6403	Constraint solving	2	This course will attempt to study the underlying techniques employed by modern day constraint solvers. In particular, solving techniques behind SAT -- such as chronological and non-chronological backtracking, conflict-driven clause learnin. Various encoding techniques as well as analysis of size of the encodings for MaxSAT and Pseudo-Boolean constraint solving will be studied. In addition, this course may also attempt to take a look at SMT (Satisfiability Modulo Theories) solving.	1. Selected research papers will be made available to the students by the instructor. 2. Handbook of Satisfiability edited by A. Biere, M. Heule, H. Van Mareen and T walsh. ISO Press, 2009. ISBN (print) : 978-1-58603-929-5, ISBN (online): 978-1-60750-376-7. 3. Handbook of Constraint Programming, edited by F. Rossi, P. Van Bree and T. walsh. Elsevier Science, 2006. ISBN:978-0-44452-726-4
CS6410	Software verification	3	The course may cover topics such as Hoare logic, abstract interpretation, abstraction refinement, k-induction, symbolic execution, variants of bounded model checking for sequential as well as concurrent programs such as loop bounding, context bounding and reorder bounding. Use of formal techniques for software testing and reasoning about termination can also be covered.	Selected research papers
CS6420	Topics in deep learning	3	This research-intensive course will focus on most recent developments in deep learning, with a focus on latest papers in top conferences and journals. This course will be a combination of lectures, seminars and hands-on projects.	Ian Goodfellow and Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, http://www.deeplearningbook.org/ .
CS6430	Stochastic processes in machine learning	1	Gaussian Processes : regression and classification, Inference techniques Dirichlet Processes : Dirichlet process mixture model and Hierarchical Dirichlet processes, Point processes : Inhomogeneous Poisson processes and variants.	1. Gaussian Processes for Machine learning by Rasmussen and Williams 2. Lecture notes in Bayesian non-parametrics by Peter Orbanz. 3. Selected research papers on various topics.
CS6440	Special topics in machine learning	3	This research-intensive course will focus on most recent developments in machine learning, with a focus on latest papers in top conferences and journals. This course will be a combination of lectures, seminars and hands-on projects.	•Relevant research papers and resources

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CS6450	Visual computing	3	Introduction to visual computing framework: Human visual recognition system; Sensing/Capturing images and videos; Tasks/Applications on image and videos; Challenges; Representation of visual data/information/signal; Learning for detection, recognition, classification, etc. Representation: Classical techniques, LBP, HoG, HoF, SIFT, STIP, bag-of-words; fisher vector, universal action models, dictionary representation; Deep representation, autoencoders, deep belief network, convnet models. Learning: k-nearest neighbour, support vector machine, softmax regression; neural networks. Applications: Image recognition/classification, face detection/recognition, action and activity recognition; Large-scale image/video classification and retrieval.	1. Handbook of Image and Video Processing (2nd Edition) by AI Bovik 2. Computer Vision : Algorithms and Applications by Richard Szeliski 3. Deep Learning - An MIT Press book by Ian Goodfellow and Yoshua Bengio and Aaron Courville
CS6460	Visual big data analytics	3	Visual computing framework, Representation of visual big data: low level representation, mid level representation, high level representation, deep representation of visual data; Performing analytics on visual big data: classification, clustering, regression, ranking; Tools, techniques, and platforms for visual big data analytics; Case study, project and paper presentation: Parallel/distributed implementation of feature extraction techniques, Parallel/distributed implementation of machine learning techniques, Multimedia application scenarios over distributed environment or cloud.	1. Scaling up Machine Learning: Parallel and Distributed Approaches Hardcover – 30 Dec 2011 by Ron Bekkerman (Editor), Mikhail Bilenko (Editor) , John Langford (Editor) 2. Deep Learning - An MIT Press book by Ian Goodfellow and Yoshua Bengio and Aaron Courville
CS6470	Topics in vision and learning	3	Course OutlineThis research-intensive course will focus on most recent developments in machine learning and computer vision, with a focus on latest papers in top conferences and journals. This course will be a combination of lectures, seminars and hands-on projects.	Relevant research papers

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CS6480	Causal inference and learning	2	Background: Probability, Bayesian networks, d-separation Introduction to causal inference: identifiability, ignorability, SUTVA, randomized experiments, interventions, selection bias, confounding, causal effect estimation. Potential outcomes framework: matching and propensity score models, natural experiments and regression discontinuity, instrumental variables, Direct vs total effects Causal graphs: encoding causal assumptions with graphical models, do-calculus and controlling for confounding, counterfactual and interventional logic, transportability, causal structure learning. Current topics in causal learning: causal invariance search, causality in machine learning, causal representations, causal explanation, causal discovery, algorithmic confounding.	Judea Pearl, Causality: Models, Reasoning and Inference, Cambridge University Press, 2nd edition, 2009. Judea Pearl, Madelyn Glymour and Nicholas Jewell: Causal Inference in Statistics: A Primer (Wiley Press 2016), available as an ebook from the UIC library. Hernan MA, Robins JM, Causal Inference, Boca Raton: Chapman and Hall, CRC, 2019 Tian J, Studies in Causal Reasoning and Learning, PhD Thesis, UCLA, 2002 Chen. B. and Pearl. J, Graphical tools for linear structural equation modeling. Technical Report-432, UCLA, 2015.
CS6483	Constraint Programming	3	This course will attempt to study the underlying techniques employed by modern day constraint solvers. In particular, solving techniques behind SAT -- such as chronological and non-chronological backtracking, conflict-driven clause learning. Various encoding techniques for cardinality constraints as well as analysis of size of the encodings for MaxSAT and Pseudo-Boolean constraint solving will be studied. In addition, this course may also attempt to take a look at SMT (Satisfiability Modulo Theories) solving. (Note: Some topics may be added/deleted to suit specific offerings of the course) For the practical learning of the concepts, the course would require the students to implement solvers and/or encodings either from scratch or on top of some constraint programming framework.	1. Selected research papers will be made available to the students by the instructor. 2. Handbook of Satisfiability edited by A. Biere, M. Heule, H. Van Mareen and T walsh. ISO Press, 2009. ISBN (print) : 978-1-58603-929-5, ISBN(online): 978-1-60750-376-7. 3. Handbook of Constraint Programming, edited by F. Rossi, P. Van Bree and T. walsh. Elsevier Science, 2006. ISBN:978-0-44452-726-4
CS6490	Hardware architecture for deep learning	3	Approximate computing and storage, low-precision deep-learning (DL) accelerators, FPGA-based DL accelerators, GPU-based DL accelerators, memristor-based DL accelerators, addressing memory-bottleneck in DL accelerators, deep learning on embedded system platforms, hardware-acceleration of cognitive tasks such as autonomous driving, differences in hardware requirements for DL training and inference, DL on virtual machine and containers, architectural review of some recently-proposed DL accelerators (e.g., TPU).	•Recent research papers on these topics •Reagen, B., Adolf, R., Whatmough, P., Wei, G. Y., & Brooks, D. (2017). Deep Learning for Computer Architects. Synthesis Lectures on Computer Architecture, 12(4), 1-123.

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CS6500	Statistical Programming	2	Probability and statistics Statistical measures and tests Introduction to statistical packages, such as SPSS, SAS, R, Stata, JMP. Statistical analyses using R and Python	The Elements of Statistical Learning (2nd edition) by Hastie, Tibshirani and Friedman
CS6510	Applied Machine Learning	3	Classification (Naive Bayes, k-NN, SVM, Neural Networks, Decision Trees, Logistic Regression, Ensemble Methods), Regression (Linear, Non-linear, k-NN, SVR), Clustering (k-means, DBSCAN, hierarchical), Dimensionality Reduction (PCA, MDS, Isomap), Gaussian Mixture Models, EM, Feature Selection, Model Selection and Performance Evaluation (Cross-Validation, Bootstrap, ROC), Time series analysis methods	Pattern Recognition and Machine Learning by Christopher Bishop
CS6520	Data Intelligence and Analytics	2	•Online Analytical Processing (OLAP) •Decisive analytics •Descriptive analytics •Predictive analytics •Prescriptive analytics •Data analysis in specific domains	
CS6530	Analytic Databases	2	Recommendation systems; Finding similar items using locality sensitive hashing; Frequent pattern mining; Link analysis in social network; Mining data streams; Large scale file systems & map-reduce; Data warehouses	
CS6540	Image and Video Analytics	2	•Image and video processing •Human visual recognition system •Detection/segmentation/recognition methods •Image and video classification models •Image and video analytics applications	
CS6550	Scaling to Big Data	2	•Distributed computing architecture •Parallel programming •Apache Hadoop framework •MapReduce programming	

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CS6560	Social media analytics	2	•What is social network? •Network models •Network centrality measures •Finding network communities •Information diffusion •Contagion and opinion formation	
CS6570	Data Acquisition and Productization	2	•Data preprocessing: extraction, cleaning, annotation, integration •Information visualization •Dashboards, Android and iOS apps. •Internet of things	
CS6650	Information Retrieval	2	•Storing, indexing and querying document data •Scoring, term weighting and the vector space model •Text classification problem and Naive Bayes •Probabilistic information retrieval •Link analysis	
CS6660	Mathematical Foundations of Data Science	3	Matrices, Vectors and Properties; Vector Spaces, Norms, Basis, Orthogonality; Matrix Decompositions: Eigen decomposition, Singular Value Decomposition; Differential Calculus: Derivatives and its significance, Partial derivatives; Optimization of single variable and multiple variable functions: Necessary and sufficient conditions; Real problems as optimization problems: Formulation and analytical solutions; Finding roots of an equation: Newton Raphson Method; Optimization via gradient methods; Probability basics, density function, counting, expectation, variance, independence, conditional probability, Poisson process, recurrences, Markov chains	
CS6670	Topics in Data Mining	3	Data Preprocessing, Data Warehousing & OLAP, Mining Frequent Patterns and Associations, Classification, Cluster Analysis, Mining Complex Types of Data (Sequence Data, Graphs, Social Networks, etc.), Text Mining, Stream Data Mining	Jiawei Han, Micheline Kamber, Jian Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2011 Yizhou Sun and Jiawei Han, Mining Heterogeneous Information Networks: Principles and Methodologies, Morgan & Claypool, 2012

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CS6680	Parallel Programming for Practitioners	1	Mutual Exclusion; Concurrent Objects; Consistency condition for Concurrent Objects: Linearizability; Concurrent Data-Structures: Sets, Queues and Stacks;	Texts: The Art of Multiprocessor Programming. Maurice Herlihy, Nir Shavit References: Multithreaded, Parallel, and Distributed Programming. Gregory R. Andrews Introduction to Parallel Computing (2nd Edition). Ananth Grama , George Karypis, Vipin Kumar, Anshul Gupta An Introduction to Parallel Programming. Peter Pacheco
CS6713	Scalable Algorithms for Data Analysis	3	In this course, we will consider some of the algorithmic aspects for addressing such challenges. The main topics covered in this course include the framework of streaming algorithms, use of approximation and randomization for designing succinct sketches, some of the well-known streaming algorithms, handling high-dimensional data, the curse of dimensionality, embedding, searching nearest-neighbors in high-dimensional data, locality sensitive hashing and large scale networks. The contents would be from some of the standard text books and literature.	- Rajaraman, Anand, and Jeffrey David Ullman. "Mining of massive datasets". Cambridge University Press, 2011. - Aggarwal, Charu C., ed. "Data streams: models and algorithms". Vol. 31. Springer Science & Business Media, 2007. - Muthukrishnan. "Data streams: Algorithms and applications." Foundations and Trends in Theoretical Computer Science, 2005. - Blum, Avrim, John Hopcroft, and Ravindran Kannan. "Foundations of data science.", 2016.
CS6843	Compilers for machine learning	3	In this course, we will study how compilers can be applied for Machine Learning applications on various parameters like design, algorithms, engineering, scalability. On the theoretical aspects: this course will study the polyhedral model theory and the various issues that could arise in dependence analysis, scheduling, and code-generation. On the implementation front, we will study the algorithmic issues when scaling to large applications.	1. Scheduling and Automatic Parallelization. Alain Darte, Yves Robert and Frédéric Vivien, Birkhäuser, New York, ISBN 0-8176-4149-1. 2. Compilers for Machine Learning Workshop: 2020 https://www.c4ml.org/ 3. The Tensor Comprehensions framework https://research.fb.com/announcing-tensor-comprehensions/ 4. Research papers as needed
CS6863	Compilers for machine learning	1	Presently, many organizations (including large organizations like Google, Facebook, Apple, etc.) and more than a dozen startups are designing efficient compilers for specialized hardware using mathematical models like polyhedral compilation techniques. Mainstream compilers like LLVM are adding explicit support to machine learning (like the Machine Learning Intermediate Representation MLIR). In this course, we will study some of these systems.	1. Compilers for Machine Learning Workshop https://www.c4ml.org/ 2. Various DSLs/frameworks. MLIR (Google), Tensor Comprehensions (FB), RStream (Reservoir), Latte (Intel), Diesel (NVIDIA), etc. 3. The Tensor Comprehensions framework https://research.fb.com/announcing-tensor-comprehensions/ 4. MLIR Primer: A Compiler Infrastructure for the End of Moore's Law: Chris Lattner et al

Course Code	Course Name	Credits	Syllabus	References
CS6870	Surveillance video analytics	3	Home/public video surveillance using deep learning, including motion detection and classification, scene understanding, event detection and recognition, people analysis, object tracking and segmentation, anomaly detection, active video understanding, human computer/robot interaction, behavior recognition, crowd analysis, fusion of vision with other sensing modalities.	Research papers
CS6880	Multimedia content analysis	3	Multimedia content analysis refers to the computerized understanding of the semantic meanings of a multimedia document, such as a video sequence with an accompanying audio track. As we are in the digital multimedia information era, tools that enable such automated analysis are becoming indispensable to be able to efficiently access, digest, and retrieve information. Information retrieval, as a field, has existed for some time. Until recently, however, the focus has been on understanding text information, e.g., how to extract key words from a document, how to categorize a document, and how to summarize a document, all based on written text. With a multimedia document, its semantics are embedded in multiple forms that are usually complimentary of each other.. It is necessary to analyze all types of data: image frames, sound tracks, texts that can be extracted from image frames, and spoken words that can be deciphered from the audio track. This usually involves segmenting the document into semantically meaningful units, classifying each unit into a predefined scene type, and indexing and summarizing the document for efficient retrieval and browsing. In this course, we discuss recent advances in using audio and visual information jointly for accomplishing the above tasks. We will describe audio and visual features that can effectively characterize scene content, present selected algorithms for segmentation and classification, and review some testbed systems for video archiving and retrieval. We will also briefly describe audio and visual descriptors and description schemes that are being considered by the MPEG-7 standard for multimedia content description	Research papers
CS6890	Fraud Analytics using predictive and social network techniques	3	Data collection, sampling, and preprocessing Descriptive analytics for fraud detection Predictive analytics for fraud detection Social network analytics for fraud detection Post processing of fraud analytics	Fraud Analytics Using Descriptive, Predictive, and Social Network Techniques: A Guide to Data Science for Fraud Detection (Wiley and SAS Business Series)

Course Code	Course Name	Credits	Syllabus	References
CS6903	Network Security	3	Defining network security, risks, threats Classifying attacks and attack vectors A brief introduction to cryptography and PKI Internet and LAN security IEEE 802.1X Standard for Access Control in networks DNS attacks, DDoS attack , port scanning, spoofing attacks, packet sniffing, and penetration testing Hands-on with Attacking tools (nmap, hping, ettercap, etc.) Preventing and mitigating risks and attacks: operational approaches DNS security, IPSec, HTTPS, Firewalls Cyber crime, botnets, and intrusion detection systems Hands-on/lab assignments and projects using wireshark and Kali Linux	Cryptography and Network Security, William Stallings, Pearson, 7th Edition, 2016, William Stallings. Network Security Essentials: Applications and Standards, William Stallings, Prentice Hall, Fifth Edition, 2013. Attacking Network Protocols, James Forshaw, 2017